

Pinned group summary: $SO(n=4, q=2)$

Pinning-Sympy automated output

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1 Basic attributes

Group	$SO(n=4, q=2)$
Matrix size	4×4
Rank	2

1.1 Symmetric bilinear form

Dimension	4
Witt index	2
Primitive element	N/A
Epsilon	N/A

Matrix:
$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} & 0 & -x_{12} \\ x_{10} & x_{11} & x_{12} & 0 \\ 0 & -x_{30} & -x_{00} & -x_{10} \\ x_{30} & 0 & -x_{01} & -x_{11} \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} \end{bmatrix}$$

1.4 Trivial characters

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	A1_x_A1
Reduced	True
Simply laced	True
Number of roots	4

2.1 Dynkin diagram



2.2 Irreducible components

This is a reducible root system with the following components:

Dynkin type	Rank
A1	1
A1	1

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
$(-1, -1, 0, 0)$	$(-1, -1, 1, 1)$	No	—	No
$(-1, 1, 0, 0)$	$(-1, 1, 1, -1)$	No	—	No
$(1, -1, 0, 0)$	$(1, -1, -1, 1)$	Yes	+	No
$(1, 1, 0, 0)$	$(1, 1, -1, -1)$	Yes	+	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

No pairs of roots generate positive integer linear combinations.